

Two's complement

Unsigned numbers only go one way

Sign bit, negate, range

- At a fixed width, the top bit is the sign
- All ones is minus one
- To negate, invert every bit and add one
- The range is not symmetric
- That asymmetric floor is easy to forget when you “just add one more.”

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Two's complement lab

Explore **signed range**, **negate** (invert + 1), and **wrap** when values leave the representable window.

Starter example: Width 8, pattern for +5 (0000_0101). Negate to see invert+1 → −5. Try min/max and a value that wraps.

Load starter example

Challenges 0 / 22 cleared

Quiz: MSB: In two's complement, the MSB is the? Answer: sign

Answer

Show hintCheckNextIdle

Quiz: MSBQuiz: all onesQuiz: negateQuiz: 8-bit minStarter +5Negate +5Set −1Set minSet max

Wrap 200Width 4 −1Quiz: 8-bit maxInvert +5Toggle MSBNegate min+1 from −1Quiz: 4-bit range

Encode −516-bit −1Quiz: same bits−1 from 0Full story

Core ideas

Signed range

Width w : from $-2^{(w-1)}$ to $2^{(w-1)}-1$.

Negate

Bitwise invert, then add 1 (mod 2^w). All ones is -1 .

Two's complement starter

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Workbook practice

- In the workbook track, write width eight on paper
- Start from plus five: bits ending in zero one zero one
- Negate by hand, invert, then add one, and confirm you land on minus five
- Then write the signed meanings of all ones, the minimum pattern with only the sign bit set
- Name one RTL pitfall: treating a signed wire as unsigned, or forgetting the asymmetric min

Pitfalls to watch

- Do not read the most significant bit as “just another magnitude bit” when the type is
- Do not expect plus one hundred twenty-eight to fit in eight-bit two’s complement, it wraps
- And remember: the browser lab is literacy
- Waveforms and RTL still demand the same signed-versus-unsigned discipline

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, negate one value by hand and list min, max, and minus one for one width
- When you are ready, take the short quiz, then continue to overflow and wrap